



SIMATS
ENGINEERING



SIMATS
Sreeveetha Institute of Medical and Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA15 Cloud Computing and Big Data Analytics

Theory Syllabus

Unit	Title	Theory Syllabus	CO
I	Cloud and Services	Cloud Computing Introduction: definition, characteristics, and benefits of cloud computing. Evolution of Cloud Computing: hardware evolution, Internet software evolution, and enabling technologies. Server Virtualization: concepts, need for virtualization, and role in cloud computing. Web Services Overview: SOAP, REST, and service-oriented communication. Cloud Service Models: IaaS, PaaS, SaaS, and XaaS.	CO1
II	Virtualization Fundamentals	Virtualization Architecture: components and layers of virtualization. Hypervisors: Type-1 and Type-2 hypervisors and comparative analysis. Security Issues in Virtualization: VM isolation, attacks, and mitigation strategies. Cloud Datacenter Architecture: physical and virtual infrastructure, resource pooling. Software Defined Datacenter: compute, storage, and network virtualization. Service-Oriented Architecture, Cloud Federation, Identity, Presence, and Privacy.	CO2
III	Access to Cloud	Cloud Infrastructure: hardware and infrastructure components. Client and Network Security: secure access mechanisms and threats. Accessing the Cloud: platforms, web applications, web APIs, and browsers. Cloud Storage Overview: object, block, and file storage models. Storage Area Network architecture. Serverless Computing. Edge and Fog Computing. Collaboration in the Cloud. Cloud Application Development and deployment.	CO3
IV	Fundamentals of Big Data	Introduction to Big Data: motivation and evolution. Distributed File Systems: architecture and importance. Data Types: structured, semi-structured, and unstructured data. Big Data Characteristics: Volume, Velocity, Variety, Veracity, and Value. Drivers for Big Data. Big Data Analytics concepts and processing models. Data Appliances and Integration Tools: ETL and data ingestion tools. Big Data Security, challenges, and applications.	CO4
V	Hadoop and MapReduce Architecture	Apache Hadoop: overview and ecosystem components. Hadoop Distributed File System: architecture, concepts, and interfaces. Data Movement in Hadoop: data ingestion and export mechanisms. MapReduce Framework: programming model and architecture. MapReduce Execution: input, output, and job lifecycle. Anatomy of a MapReduce Job Run. Failures in Classical MapReduce. YARN, scheduling, serialization, and analytics output.	CO5

Reference Textbooks

No.	Reference Textbook
1	Rajkumar Buyya, Christian Vecchiola, and S. Thamarai Selvi, Mastering Cloud Computing: Foundations and Applications Programming, Morgan Kaufmann, 2013.
2	Kai Hwang, Geoffrey C. Fox, and Jack J. Dongarra, Distributed and Cloud Computing: From Parallel Processing to the Internet of Things, Morgan Kaufmann, 2012.
3	Thomas Erl, Ricardo Puttini, and Zaigham Mahmood, Cloud Computing: Concepts, Technology and Architecture, Prentice Hall, 2013.
4	Tom White, Hadoop: The Definitive Guide, 4th Edition, O'Reilly Media, 2015.
5	Viktor Mayer-Schonberger and Kenneth Cukier, Big Data: A Revolution That Will Transform How We Live, Work, and Think, John Murray, 2013.

Lab Programs

No.	Lab Program	Guided Project Title
1	Create a simple cloud software application and provide it as a service using any cloud service provider to demonstrate Software as a Service.	Guided Project 1: SaaS Application Builder Foundation
2	Create a simple cloud software application for Flight Reservation System using any cloud service provider to demonstrate SaaS.	Guided Project 2: Flight Reservation SaaS Workflow
3	Create a simple cloud software application for Property Buying and Rental process in Chennai using any cloud service provider to demonstrate SaaS.	Guided Project 3: Property Rental Cloud Service Portal
4	Create a simple cloud software application for Car Booking Reservation System using any cloud service provider to demonstrate SaaS.	Guided Project 4: Car Booking SaaS Reservation System
5	Create a simple cloud software application for SIMATS Library Book Reservation System using any cloud service provider to demonstrate SaaS.	Guided Project 5: SIMATS Library Reservation Cloud App
6	Create a simple cloud software application for product selling using any cloud service provider to demonstrate SaaS.	Guided Project 6: Product Selling SaaS Storefront
7	Demonstrate virtualization by installing a Type-2 hypervisor and creating/configuring a Windows or Linux VM image.	Guided Project 7: Type-2 Hypervisor Installation and VM Image Setup
8	Create a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage disk using Type-2 virtualization software.	Guided Project 8: Resource-Constrained Virtual Machine Builder
9	Create a virtual hard disk and allocate storage using VMware Workstation.	Guided Project 9: Virtual Disk Provisioning and Storage Allocation
10	Create a snapshot of a VM and test it by loading the previous version.	Guided Project 10: VM Snapshot and Restore Lifecycle
11	Create a clone of a VM and test it by loading the cloned VM.	Guided Project 11: Virtual Machine Cloning and Validation
12	Change hardware compatibility of an already created and configured VM.	Guided Project 12: VM Hardware Compatibility Migration
13	Demonstrate IaaS by creating a resource group using Azure and configuring minimum CPU, RAM, and storage.	Guided Project 13: Azure IaaS Resource Group Foundation
14	Demonstrate IaaS by creating a virtual machine using Azure and configuring required memory and CPU.	Guided Project 14: Azure Virtual Machine Provisioning
15	Demonstrate IaaS by establishing remote connection, launching the created VM image, and running it on desktop.	Guided Project 15: Remote Cloud VM Access and Launch

No.	Lab Program	Guided Project Title
16	Demonstrate PaaS by creating and configuring a new VM image in any public cloud service provider.	Guided Project 16: Public Cloud PaaS Environment Setup
17	Create a simple web application using Java or Python and host it in Azure, GCP, or AWS to demonstrate PaaS.	Guided Project 17: Cloud-Hosted Java/Python Web Application
18	Demonstrate Storage as a Service by creating and configuring storage in a public cloud environment.	Guided Project 18: Storage-as-a-Service Configuration
19	Create a storage service using Azure, GCP, or AWS and check public accessibility of the stored file.	Guided Project 19: Controlled Cloud Storage Access Test
20	Demonstrate Database as a Service by creating and configuring a database service in a public cloud environment.	Guided Project 20: Database-as-a-Service Setup
21	Create a SQL storage service and perform a basic query using Azure, GCP, or AWS to demonstrate DaaS.	Guided Project 21: Cloud SQL Query Execution
22	Perform basic configuration setup for installing Hadoop 2.x by creating hduser and SSH localhost.	Guided Project 22: Hadoop User and SSH Foundation
23	Install Hadoop 2.x and configure the NameNode and DataNode.	Guided Project 23: Hadoop NameNode and DataNode Configuration
24	Launch Hadoop 2.x and test the MapReduce platform with Hadoop.	Guided Project 24: Hadoop MapReduce Platform Test
25	Launch Hadoop 2.x and perform a MapReduce program for a word count problem.	Guided Project 25: MapReduce Word Count Analytics